

Surname

Forename(s)

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/2

PAPER 2 Theory

JUNE 2020 SESSION

2 hours

Candidates answer on the question paper

Additional materials: Calculator (Optional)

Allow candidates 5 minutes to count pages before the examination.**The Periodic Table is provided on page 16.**

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Ask the invigilator for a replacement if there are missing pages.

Write your answers in the spaces provided on the question paper.

Section A

Answer **all** questions.

Section B

Answer any **two** questions.

Section C

Answer any **two** questions.

Section D

Answer any **two** questions.

FOR EXAMINER USE

SECTION A

B7

B8

B9

C10

C11

C12

D13

D14

D15

TOTAL

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of 16 printed pages.

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[Turn over

Section A

Answer *all* the questions in this section in the spaces provided.

(a)

Fig. 1.1 shows a human tooth.

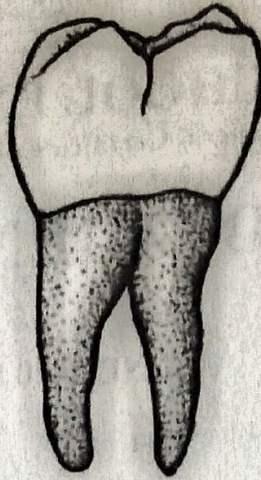


Fig. 1.1

(i) Name the tooth shown.

.....

[1]

ii) State the function of the tooth.

.....

.....

[1]

(b)

Explain the importance of mechanical digestion.

.....

.....

.....

[2]

(c)

State the end product(s) of the digestion of

(i) proteins,

.....

[1]

(ii) fats.

1

.....

2

.....

[2]

2.

(a)

3

State any **one** natural method of contraception.

.....
.....
.....

[1]

(b)

Describe how the pill prevents pregnancy.

.....
.....
.....

[3]

(c)

State any **two** signs/symptoms of chancroid in males.

.....
.....
.....

[2]

Fig 3.1 shows the blast furnace used in the extraction of iron.

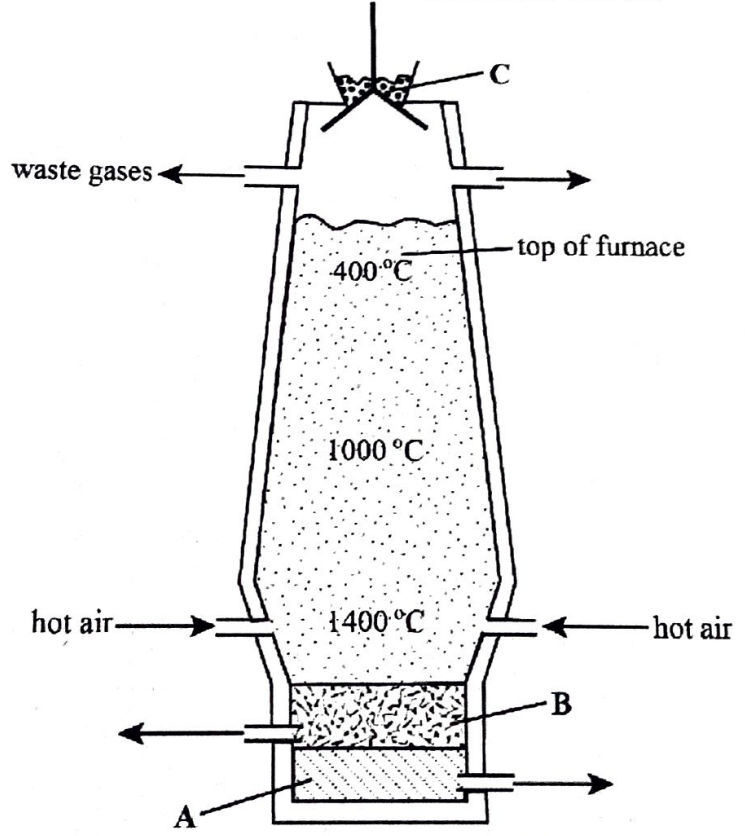


Fig 3.1

(a)

(i)

Identify substances A and B.

A
.....
B
.....

[2]

(II) C is a mixture of three raw materials.

State any **one** raw material in the mixture.

[1]

(b)

Describe the reaction which occurs near the top of the furnace.

[3]

4.

Nitrogen gas and hydrogen gas react in a reversible reaction to produce ammonia gas.

(a)

Define the term *reversible reaction*.

[2]

(b)

Write a balanced chemical equation for the production of ammonia gas.

[2]

(c)

(i)

State the **three** conditions needed for the maximum yield of ammonia.

1

2

3

[3]

(ii)

Explain how any **one** of the conditions named in (i) leads to an increase in the yield of ammonia.

[1]

5.

(a)

Define the term

(i)

momentum.

[1]

(ii)

inertia.

[1]

(b)

State the difference between mass and weight.

[1]

(c)

State Newton's first law of motion.

[1]

(d)

A ball of mass 0.4 kg accelerates uniformly at 2 m/s^2 .

Calculate the force of the ball.

[3]

6. (a)

Fig.6.1 shows power demand between 6 am and 4 pm at a boarding school which uses only electricity as its source of energy.

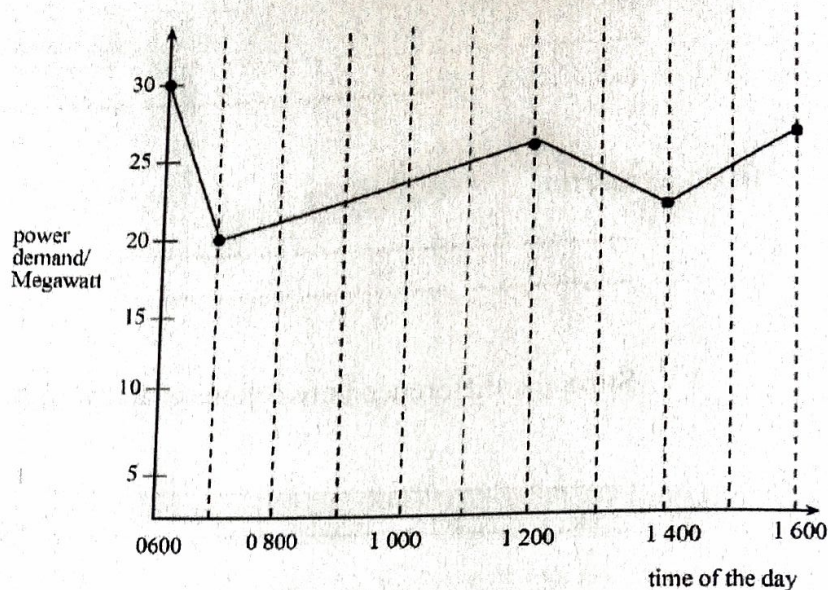


Fig.6.1

- (i) Use **Fig.6.1** to identify the time of the day with the highest power demand.

[1]

- (ii) The meal times for the boarding school are:

breakfast at 0700

lunch at 1300

supper at 1700

Explain why there is a high power demand at 1200.

[2]

- (iii) Describe how the school may reduce the amount of electricity they use per day.

[1]

(b)

State the SI units of mass and power.

mass

power

Section B

Answer any **two** questions in this section in the spaces provided.

7.

(a)

Write a word equation for photosynthesis.

[4]

(b)

State **one** use for each of the products of photosynthesis.

.....

.....

.....

[2]

(c)

Fig.7.1 shows the internal section of a leaf.

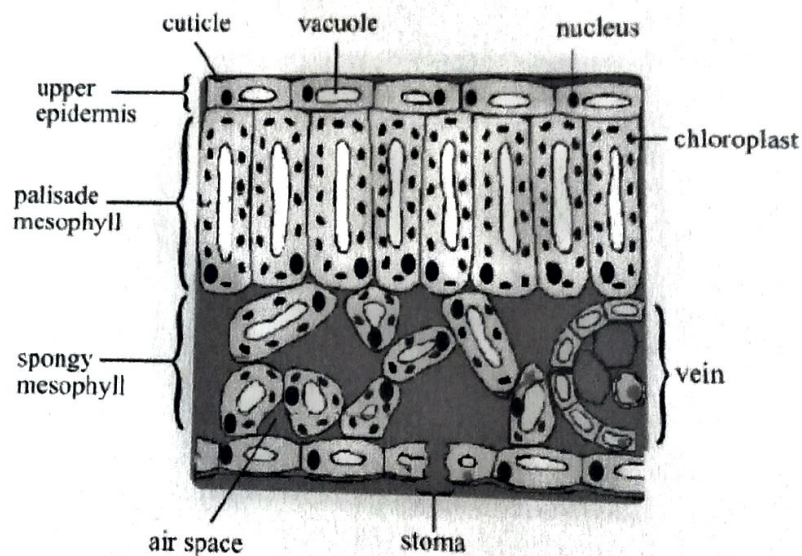


Fig.7.1

Explain how the leaf is adapted to maximise the rate of photosynthesis.

.....

.....

.....

.....

.....

8.

(a)

Define *aerobic respiration*.

.....

.....

.....

[3]

(b)

A boy developed muscle fatigue while taking part in a soccer match.

Explain what caused the muscle fatigue.

.....

.....

.....

.....

[4]

(c)

Give any **three** examples of artificial ecosystems.

.....

.....

.....

[3]

9.

(a)

Name any **two** body fluids through which the HIV virus can be transmitted.

.....

.....

[2]

(b)

Describe how the mosquito can be destroyed at any **two** stages of its life cycle.

.....

.....

.....

.....

.....

.....

[4]

(c)

(i)

Describe any **two** signs/symptoms of cholera.

.....

.....

[2]

(ii)

Explain any **one** method used in the treatment of cholera.

.....

.....

[2]

Section C

Answer any two questions in this section in the spaces provided.

10. (a) (i) State the type of bonding in magnesium oxide (MgO).
..... [1]
- (ii) Describe how magnesium oxide can be prepared.
.....
..... [2]
- (b) Describe, giving a reason, why magnesium oxide does not react with copper.
.....
..... [2]
- (c) (i) Calculate the molecular mass of magnesium oxide.
..... [2]
- (ii) Calculate the percentage of oxygen in magnesium oxide.
..... [3]
11. (a) Calcium sulphate can be formed from the reaction between calcium carbonate and solution X.
Two other substances are produced during the reaction.
- (i) Name solution X.
..... [1]
- (ii) Give the other two products.
1
2

- (iii) Name the type of reaction that occurs between calcium carbonate and solution X.

.....

[1]

- (iv) To prepare a pure sample of calcium sulphate, excess calcium carbonate has to be added to the solution X.
Explain why the calcium carbonate should be in excess.

.....

[1]

- (v) Name the process that is used to separate the excess calcium carbonate from the calcium sulphate solution.

.....

[1]

- (b) The strength of an acid or a base can be determined using the pH scale.

- (i) State the pH values on the pH scale.

.....

[1]

- (ii) Give the pH value that represents a neutral substance.

.....

[1]

- (c) List any **two** properties of alkaline substances.

1

2

.....

[2]

12. (a)

Dyes in black ink are separated using a strip of a filter paper and a solvent. The ink is placed on the filter paper and the position of the ink is marked in pencil. The filter paper strip is dipped into the solvent, ensuring that the solvent is just below the ink.

(i) Name the process used to separate the dyes in the black ink.

[1]

(ii) Suggest, with a reason, a suitable solvent.

solvent

reason

[2]

(iii) Explain why the position of the ink is marked in pencil and not in ink.

.....
.....

[2]

(iv) State two properties of dyes which make it possible to separate them.

.....
.....

[2]

(v) Name the force that enables the solvent to move up the filter paper.

.....

[1]

(b) State a method which can be used to separate

(i) ethanol from water,

.....

[1]

(ii) iron filings from sulphur.

.....

[1]

Section D

Answer any two questions in this section in the spaces provided.

(a)

A polythene rod can be charged negatively by rubbing it with a woollen cloth.

Describe how the rod becomes negatively charged through the rubbing done.

(3)

(b)

A man drags a 60 kg bag up an inclined plane of length 4.5 m and height 1.5 m to load it into a lorry.

(i)

Calculate the velocity ratio, VR, of the inclined plane.

(ii)

State the effect, on the velocity ratio of the inclined plane, of increasing the height.

(c)

State, giving a reason, the method by which heat is transferred from the sun to the earth.

(d)

The heat from the sun can be used for heating using a solar cooker. State any one feature of the cooker which ensures that maximum heat is focussed on one point (where the pot is placed).

14. (a) Table 14.1 shows Tobias' record of electricity use for two appliances used in one month.

Table 14.1

appliance	casing material	power rating/W	total time switched on/hour
radio	plastic	500	60
hot plate	metal	2000	30

- (i) State, giving a reason, which one of the two appliances has a two-pin plug.

.....
 [2]

- (ii) Electricity costs 10 c per unit.

Calculate the cost of using the hot plate for the month.

[3]

- (b) State any **three** precautions which should be observed when using electricity.

.....

 [3]

- (c) Give any **two** ways of saving electricity.

.....
 [2]

15.

(a)

Telecommunication messages can be transmitted in a variety of ways by a wide range of devices which include cell phones.

- (i) Name any other **one** device that can be used to send messages apart from a cell phone.

.....
.....

[1]

- (ii) State the energy conversion that occurs in the device named in (i).

.....
.....

[2]

- (iii) State any **two** types of information that can be transmitted by a cell phone.

1

2

.....
.....

[2]

(b)

(i)

Coaxial cables are one type of media for signal transmission.
State any other **two** types of media for signal transmission.

1

2

.....
.....

- (ii) Describe how signal is transmitted by coaxial cables.

.....
.....
.....
.....

DATA SHEET

The Periodic Table of the Elements

Group																																																																			
I	II											III	IV	V	VI	VII	O																																																		
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*58-71 Lanthanoid series
†90-103 Actinoid series

Key
 $\frac{a}{b} X$
 a = relative atomic mass
 X = atomic symbol
 b = proton (atomic) Number

140 Ce Cerium	141 Pr Praseodymium	144 Nd Neodymium	150 Pm Promethium	152 Sm Samarium	157 Eu Europium	162 Gd Gadolinium	158 Tb Terbium	162 Dy Dysprosium	165 Ho Holmium	167 Er Erbium	169 Tm Thulium	173 Yb Ytterbium	175 Lu Lutetium
88 Th Thorium	89 Pa Protactinium	90 U Uranium	92 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

The volume of one mole of any gas is 28 dm³ at room temperature and pressure (r.t.p.)